

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To help farmers choose the most suitable crop and optimize agricultural production using machine learning and real-time data analysis.
Scope	The project covers soil data analysis, weather data integration, crop recommendation, yield prediction, and a user-friendly web/mobile platform for farmers.
Problem Statement	
Description	Farmers face difficulty in selecting the right crop due to lack of soil analysis, unpredictable weather, and limited access to expert guidance. This leads to low yield, wastage of resources, and reduced income.
Impact	Incorrect crop selection causes low productivity, financial loss, excessive use of fertilizers and water, and unsustainable farming practices.
Proposed Solution	
Approach	The system will use machine learning algorithms trained on soil, weather, and crop datasets to recommend suitable crops and predict yield. It will provide actionable insights to improve productivity and sustainability.
Key Features	• Soil nutrient analysis (N, P, K, pH, etc.) • Real-time weather data integration • Crop suitability recommendation • Yield prediction • Fertilizer & water usage optimization • User-friendly dashboard for farmers • Multi-language support for rural users

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	Server / Cloud resources for model training and deployment	AWS EC2 (4 vCPU, 16 GB RAM)
Memory	System memory for smooth execution	16 GB RAM
Storage	Storage for datasets, models, and logs	500 GB SSD
Software		
Frameworks	Web framework	Django (Python Framework)
Libraries	Machine learning & data processing libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE / Tools	VS Code, Jupyter Notebook, Postman
Data		
Data	Datasets for training and testing	Soil Nutrient Dataset, Crop Dataset, Weather Dataset (OpenWeather API)